

AI Usage DISCLOSURE FORM

1. Team Details

Team Name: Ember

Project / Product Name: Sentry

Organization / Institution (if any): Vellore Institute of Technology, Vellore

Submission Date: 30 September 2026

2. AI Usage Declaration

Did your team use any Artificial Intelligence (AI) in developing this project? Yes

3. Purpose of AI Usage (Brief Details)

Idea generation / brainstorming: Done by the team. System architecture, feature set, and the core product differentiators (verified cache with opposite-action detection, self-critique with a proven latency guarantee, honest failure handling) were conceived and directed by the team

Code generation or assistance. Done with the help of AI. The team specified exact technical requirements and implementation details; AI tools (Claude Code, Google Antigravity) generated code from those specifications, which the team then tested, debugged, and corrected

UI / UX design: visual direction and design requirements were specified by the team and the implementation was AI-assisted

Content creation: Project documentation (README, technical evaluation report) was drafted with AI assistance based on team-specified content and reviewed for accuracy. Multiple reviews and edits were made by team. Use of AI to make the README neater and more understandable.

Data analysis: The team designed the evaluation methodology (3-way architectural ablation, confidence calibration study) and directed the analysis; AI assisted in running and summarizing the results

Testing / debugging: The team independently ran live tests, identified multiple real bugs through direct system testing (e.g. an off-domain hallucination issue, a latency-SLA

regression from a failover interaction), and directed the specific fixes, which were then re-verified

Other: Comparative research against prior hackathon submissions to benchmark technical approach, understanding their project repositories, their architecture and features were done with AI assistance.

4. Feature Origin Classification (Please add/delete based on the number of the feature)

1. Feature Name: Core troubleshooting pipeline (query enrichment, structured extraction, deeplink ordering)

Self-Generated/AI-Generated/Both: Both

Description: Tools: Claude, Claude Code, Google Antigravity. The team designed the full pipeline architecture, the schema contract, and the safety constraints, then specified precise implementation requirements. AI generated code to those specifications. The team identified a real off-domain hallucination bug through live testing, directed a two-stage fix (a prompt-level instruction, then a deterministic backstop after the first proved insufficient), and verified the fix against adversarial test cases.

2. Feature Name: Verified semantic cache with opposite-action veto

Self-Generated/AI-Generated/Both: Both

Description: Tools: Claude Code. The team specified the core requirement, that paraphrased queries with opposite meaning (e.g. 'turn on' vs 'turn off') must never share a cached answer, a failure mode most naive semantic caches miss. Implementation was AI-generated; the team independently audited the resulting code for correctness before merging.

3. Feature Name: Hybrid retrieval (BM25 + label matching)

Self-Generated/AI-Generated/Both: Both

Description: Tools: Claude Code. The team ran live testing against the real 578-entry catalog, identified specific retrieval misses, diagnosed the root cause, and directed a targeted fix. Results were independently verified: 94-100% accuracy across three held-out test sets.

4. Feature Name: Dual-key failover, self-critique pass, off-domain guard, one-action-one-screen validator

Self-Generated/AI-Generated/Both: Both

Description: Tools: Claude, Claude Code. Each feature was initiated by the team after identifying a real operational risk (API quota exhaustion during testing, hallucinated diagnoses, off-topic queries, inconsistent action granularity). The team specified exact fix requirements, including a mathematically proven latency ceiling for self-critique, then found and directed the correction of a subtle regression where two independently-correct features interacted badly.

5. Feature Name: Demo frontend (demo/index.html)

Self-Generated/AI-Generated/Both: Both

Description: Tools: Claude Code, Google Antigravity. The team specified feature requirements, interaction design, and visual direction (including a full redesign pass), then tested and gave iterative feedback across multiple rounds before finalizing.

6. Feature Name: Evaluation suite (3-way ablation, calibration analysis, smoke tests)

Self-Generated/AI-Generated/Both: Both

Description: Tools: Claude Code. The team designed the evaluation methodology directly from the hackathon's own evaluation criteria, including a confidence-calibration analysis that surfaced an honest ceiling effect rather than hiding it. Results were independently verified by running the suite and inspecting real output.

5. Ethical & Compliance Confirmation

AI usage complies with guidelines and policies. **Yes**

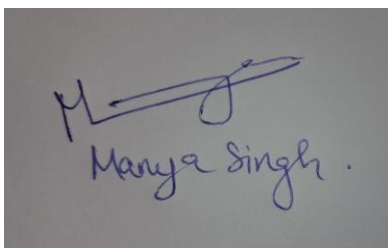
No proprietary or copyrighted data misused. **I Agree**

6. Declaration & Sign-Off

Name of Team Representative: Manya Singh

Role: Team Lead / Backend & Integration

Signature: Manya Singh

A photograph of a handwritten signature in blue ink on a grey background. The signature is stylized, starting with a large 'M' and ending with a long horizontal stroke. Below the signature, the name 'Manya Singh' is written in a cursive script.

